

Stage 1: Rollout Sampling

Query x

$y(1)$ ✓

$y(2)$ ✗

$y(3)$ ✓

$y(4)$ ✗

$y(5)$ ✗

$y(6)$ ✓

$\mathcal{G}^+$

$\mathcal{G}^-$

Verifier $\rightarrow$ partition

Stage 2: Contrastive Token-Level Signal

positive hint
 y_c^*

K negative hints
 $y_w^* \sim \mathcal{G}^- / \{y\}$

$$\log \pi_T(y_t | x, y_c^*, y_{<t})$$

$$\log \frac{1}{K} \sum_k \pi_T(y_t | x, y_w^*, y_{<t})$$

$e_{ctr,t}$
style drift cancels

Cancellation property

Identical prompt template for y_c^* and y_w^* ; shared stylistic component cancels in subtraction.

Special Design

1. Using **K-marginalization** for negative hints
2. **Not to use the target rollout itself**

Stage 3: Verifier-Anchored Modulation & Loss

$$r_t = \lambda \cdot \tanh(e_{ctr,t} / \tau)$$

$$\tilde{A}_t = \text{same_sign_clamp}(A_{ORM} + r_t)$$

verifier anchors direction

$$m_t = \mathbb{1}[|r_t| > \delta]$$

Two-Path Loss: L_{RLCSD}

$$m_t = 0$$

$L_{PPO-clip}(A_{ORM})$
unmodulated path

$$m_t = 1$$

$\eta \cdot L_{PPO-clip}(\tilde{A}_t)$
modulated path